

Bayesian Inference for High-Frequency Market Microstructure

A Hierarchical Approach Combining Bayesian Linear Regression, MC Dropout Neural Networks, and HMM Mixture of Experts on the Optiver Closing Auction Dataset

Final Project — Bayesian Analysis

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1. Introduction

Financial markets generate vast quantities of structured data in real time, and the ability to extract predictive signals from this data is of fundamental importance to market participants. This project focuses on one of the most information-rich and time-critical episodes in equity trading: the closing auction of the NASDAQ exchange, which takes place in the final ten minutes of each trading day (15:50–16:00 EST). During this window, market participants submit buy and sell orders that accumulate into an order book, and the exchange determines a single clearing price that maximises the matched volume.

The dataset used in this work is provided by Optiver and covers 200 NASDAQ-listed stocks across 481 trading days, yielding approximately 5.2 million observations. Each row represents the state of the order book for a specific stock at a specific 10-second interval within the auction window. The task is to predict the 60-second forward return of each stock relative to the synthetic market index — a formulation that imposes a structural zero-sum constraint on predictions and makes cross-sectional signal the primary source of predictive power.

Standard machine learning approaches to this problem treat it as a regression task and optimise a single point estimate. This approach has a fundamental limitation in financial applications: it provides no measure of the model's confidence in its predictions. A model that produces a point estimate of +2.4 basis points cannot distinguish between a case where it is highly confident (e.g., strong order book imbalance with corroborating signals) and one where it is highly uncertain (e.g., conflicting signals near a market regime transition). This distinction is critical for risk-adjusted position sizing and portfolio construction.

The Bayesian framework addresses this limitation directly. By treating model parameters θ as random variables with a prior distribution $p(\theta)$, and updating this prior with observed data D via Bayes' theorem:

$$p(\theta \mid D) \propto p(D \mid \theta) \cdot p(\theta)$$

the resulting posterior $p(\theta \mid D)$ encodes both the most likely parameter values and the uncertainty about them. The posterior predictive distribution $p(y^* \mid x^*, D)$ then provides, for each new observation, not just a point

estimate but a full probability distribution over possible outcomes — enabling calibrated confidence intervals and principled uncertainty quantification.

This project constructs a complete hierarchical Bayesian pipeline consisting of three complementary modelling layers. First, a Bayesian Linear Regression (BLR) model is fitted on the full 312-day training window to capture stable long-range linear signals with analytically exact uncertainty. Second, a Monte Carlo Dropout Bayesian Neural Network (BNN) is trained on the most recent 56 days to model non-linear short-term dynamics with approximate posterior inference via stochastic forward passes. Third, an inverse-variance ensemble combines both models via Bayesian Model Averaging (BMA), weighting each model by its stated confidence at each observation.

Three subsequent improvements are then layered on top: (1) extended lag and rolling features to better capture autocorrelation in financial time series, (2) within-fold uncertainty calibration to correct the systematic overconfidence of the baseline ensemble, and (3) a Hidden Markov Model (HMM) Mixture of Experts that detects latent market regimes and conditions each expert model on the identified regime.

The paper is structured as follows. Section 2 describes the dataset and exploratory analysis. Section 3 covers feature engineering across five distinct groups. Section 4 explains feature selection via BayesianRidge. Section 5 describes the walk-forward cross-validation scheme. Section 6 provides detailed derivations of all three models. Section 7 explains post-processing. Sections 8 through 12 present results and three improvements. Section 13 concludes.

2. Market Context and Problem Formulation

2.1 The Closing Auction Mechanism

The NASDAQ closing auction is not a standard continuous-trading session. During the final ten minutes of the trading day, the exchange operates in a hybrid mode: traditional limit orders continue to be submitted to the non-auction book (bid/ask book), while simultaneously auction-specific orders are submitted to compete for execution at a single clearing price. This dual structure produces two distinct information sources that the model must integrate.

The clearing price referred as the reference price in the dataset. This is the price at which the maximum number of shares can be matched between buyers and sellers. The `far_price` is a projection of this clearing price that accounts only for auction orders, while `near_price` also incorporates orders from the continuous book. The gap between these quantities, captured by the feature `price_disc_gap`, indicates how much price discovery tension exists between the two order flows.

As the auction progresses and the 16:00 closing moment approaches, urgency increases non-linearly. Large institutional investors who need to execute before market close face increasing time pressure, and their order flow tends to accelerate in the final 60 seconds. This structural property motivates several temporal features (`is_last_60`, `urgency_exp`) and explains why target volatility is 2–3× higher in the last minute than in the earlier part of the auction.

2.2 The Prediction Target and Zero-Sum Constraint

The prediction target $Y_{\{i,t,s\}}$ is defined as the stock return minus the index return over the subsequent 60 seconds:

$$Y_{\{i,t,s\}} = \left(\text{StockWAP}_{\{i,t,s+60\}} / \text{StockWAP}_{\{i,t,s\}} - \text{IndexWAP}_{\{t,s+60\}} / \text{IndexWAP}_{\{t,s\}} \right) \times 10,000$$

This formulation has several important properties. First, the unit is basis points (1 bp = 0.01% price move), a standard unit in financial trading. Second, the target is a relative return — it measures how much stock i outperforms or underperforms the market index, not its absolute return. Third, and most critically, the target imposes a structural zero-sum constraint.

Since the index return is a weighted average of individual stock returns, and the target subtracts this average from each stock's return, the cross-sectional sum of all targets must be approximately zero at each moment:

$$\sum_{i=1}^N Y_{i,t,s} \approx 0 \quad \text{for all } (t, s)$$

This means a model that systematically predicts all stocks will outperform the index is structurally wrong — it is predicting market direction rather than relative cross-sectional positioning. The zero-sum adjustment (Section 7.1) enforces this constraint as a post-processing step, and the MAE+ZS metric evaluates performance after this adjustment.

The target distribution has mean -0.048 bp (near zero, consistent with the zero-sum constraint) and standard deviation 9.41 bp. The distribution is approximately symmetric when clipped at ± 30 bp, with heavy tails caused by extreme auction events.

2.3 Evaluation Metrics

Metric	Formula	What It Measures
MAE	$\text{mean}(y_{\text{hat}} - y_{\text{true}})$	Raw prediction error in basis points. Ignores cross-sectional structure.
MAE+ZS	$\text{mean}(y_{\text{hat_zs}} - y_{\text{true}})$, $y_{\text{hat_zs}} = y_{\text{hat}} - \text{cross-sectional mean}$	MAE after zero-sum adjustment. The operationally relevant metric.
std/MAE	$\text{mean}(\text{predicted_std}) / \text{mean}(\text{error})$	Calibration quality. Target = 1.0. Below 1: overconfident. Above 1: underconfident.
MAE+ZS / MAE ratio	$\text{MAE+ZS} / \text{MAE}$	Cross-sectional neutrality. Ratio near 1.0 means predictions already satisfy zero-sum.

3. Data and Exploratory Analysis

3.1 Dataset Structure

The dataset contains 5,237,892 observations from the Optiver Trading at the Close competition. Each row represents a unique combination of (stock_id, date_id, seconds_in_bucket), where seconds_in_bucket ranges from 0 to 540 in steps of 10, giving 54 time steps per stock per day. Stock and date identifiers are anonymised integers to prevent information leakage.

Column	Type	Description
stock_id	int16	Anonymised stock identifier (0–199, 200 unique stocks)
date_id	int16	Trading day identifier (0–480, 481 unique days)
seconds_in_bucket	int16	Time in auction: 0, 10, 20, ..., 540 seconds
bid_price / ask_price	float32	Best bid and ask prices in the non-auction continuous book
bid_size / ask_size	float32	Volume available at best bid and ask (in USD)
wap	float32	Weighted average price: $(\text{bid} \times \text{ask_size} + \text{ask} \times \text{bid_size}) / (\text{bid_size} + \text{ask_size})$
imbalance_size	float32	Volume of unmatched orders at the current reference price (USD)
imbalance_buy_sell_flag	int8	Direction: +1 buy-side, -1 sell-side, 0 balanced
matched_size	float32	Volume already matched at reference price (USD)
reference_price	float32	Current clearing price if auction closed now
far_price	float32	Indicative clearing price (auction orders only). NaN before price discovery.
near_price	float32	Indicative clearing price (auction + continuous orders). NaN early on.
target	float32	60-second forward relative return in bp (training set only)

3.2 Key Observations from EDA

Three diagnostics are examined. First, the target distribution is approximately symmetric around zero when clipped at ± 30 bp, with mean -0.048 and standard deviation 9.41 bp. The near-zero mean is consistent with the zero-sum constraint: on average, no stock systematically outperforms the index.

Second, target volatility increases sharply in the final 60 seconds of the auction (seconds 480–540). This motivates the `is_last_60` binary feature and the `urgency_exp` exponential feature, which spike precisely in this window. The increase reflects institutional investors executing large orders before the market closes.

Third, `far_price` and `near_price` are absent (NaN) in the early part of the auction when price discovery has not yet begun. The `auction_active` binary feature handles this missingness explicitly, and these features carry their NaN structure forward into `price_disc_gap`.

4. Feature Engineering

Feature engineering is the most important step in this pipeline. All 36 candidate features are constructed from the raw 13 columns, organised into five groups by financial motivation. A fundamental constraint applies throughout: every `shift()`, `diff()`, and `rolling()` operation is computed strictly within `groupby([stock_id, date_id])` to prevent information leakage across stocks or across trading days. A global `df.diff(1)` would, for example, compute the change between stock *j*'s last observation on day *t* and stock *i*'s first observation on day *t*+1 — a nonsensical cross-stock quantity.

4.1 Group 1 — Microstructure Features

These features characterise the static state of the order book at a single point in time, answering the question: what is the current supply/demand balance?

Feature	Formula	Financial Interpretation
<code>imb1</code>	$(\text{ask} - \text{bid}) / (\text{ask} + \text{bid})$	Normalised bid-ask spread. Measures transaction cost and liquidity. Wider spread indicates less liquid conditions.
<code>ofi_ratio</code>	$\text{imbalance_size} \times \text{flag} / (\text{matched} + 1)$	Order Flow Imbalance. The primary alpha source. Measures unfilled directional pressure relative to matched flow. High positive OFI signals unmet buy demand that will push prices up.
<code>depth_imbalance</code>	$(\text{bid_size} - \text{ask_size}) / (\text{bid_size} + \text{ask_size} + 1)$	Supply/demand tilt at best quotes. Captures which side of the market is more aggressively queued.
<code>price_disc_gap</code>	$(\text{far_price} - \text{near_price}) / \text{reference_price}$	Price discovery tension between auction-only and combined clearing prices. Non-zero only when both prices are available.
<code>matched_ratio</code>	$\text{imbalance_size} / (\text{matched_size} + 1)$	Ratio of unfilled to filled volume. High ratio indicates demand that cannot be satisfied at current price.
<code>auction_active</code>	<code>far_price.notna()</code>	Binary indicator: has auction price discovery begun? Marks a structural transition in auction dynamics.
<code>market_urgency</code>	<code>price_spread * depth_imbalance</code>	Interaction term: wide spread combined with strong imbalance signals stressed, urgent market conditions.

4.2 Group 2 — Price Dynamics

These features capture how prices are evolving over time, providing first- and second-order information about momentum and acceleration.

The Weighted Average Price (WAP) is central here. Unlike the mid-price $(\text{bid} + \text{ask})/2$, WAP weights each side by the opposing side's volume: $\text{WAP} = (\text{bid} \times \text{ask_size} + \text{ask} \times \text{bid_size}) / (\text{bid_size} + \text{ask_size})$. This means a bid-

heavy book pushes WAP closer to the ask — reflecting where actual transactions are most likely to occur. WAP is therefore a more accurate proxy for the true market price than the simple mid-point.

Feature	Formula	Financial Interpretation
wap_velocity	wap.diff(1)	First-order WAP momentum. Direction and speed of recent price movement within the auction.
wap_acceleration	wap.diff(1).diff(1)	Second-order WAP dynamics. Positive: momentum is increasing. Negative: momentum is decelerating.
ref_velocity	reference_price.diff(1)	Rate of change in the clearing price. Captures how quickly the auction price is moving.
ref_acceleration	ref_velocity.diff(1)	Second derivative of clearing price.
matched_size_diff	matched_size.diff(1)	Change in matched volume. Spikes in the final 60 seconds as large institutional orders execute.
matched_size_vol	matched_size.rolling(5).std()	Rolling volatility of matched volume. High variability indicates erratic order flow.
wap_rolling5_std	wap.rolling(5).std()	Local price volatility over 50 seconds. Signals regions where the model should produce wider uncertainty intervals.

4.3 Group 3 — Temporal Pressure

As the auction approaches its 16:00 close, urgency increases non-linearly. These features encode this temporal structure explicitly.

Feature	Formula	Financial Interpretation
seconds_to_close	540 – seconds_in_bucket	Simple linear countdown. Useful as a baseline temporal feature.
time_log	$\log(1 + \text{seconds_to_close})$	Log-compressed time. Reduces the influence of the early, less-dynamic auction period.
urgency_exp	$\exp(-\text{seconds_to_close} / 60)$	Non-linear urgency. Near-zero for most of the auction, spikes sharply in the final 60 seconds.
is_last_60	$\text{seconds_in_bucket} \geq 480$	Binary flag for the final minute. Target volatility is 2–3× higher in this window.
urgency_time_weighted	$\text{market_urgency} \times (1 + 540/(\text{stc}+1))$	Order-book stress amplified by temporal proximity to close. Captures the intersection of price pressure and time pressure.

4.4 Group 4 — Lag and Rolling Features

Financial time series exhibit strong autocorrelation: recent events predict near-future events. These features give the model memory of recent order book dynamics.

A critical design decision applies to target-based rolling features: any feature derived from the target variable must use `.shift(1)` to ensure only past values are used. Without `shift(1)`, the rolling mean at time t includes the value `target[t]`, which is the quantity being predicted — a direct data leak. In an earlier version of this pipeline, this error reduced MAE to near zero (approximately 0.00002), falsely appearing as a perfect model. After applying `shift(1)`, MAE returned to the realistic 2.4 bp range.

Feature	Formula	Financial Interpretation
target_lag1	target.shift(1)	Return one step ago. The strongest feature ($ \text{coef} = 7.86$). Short-term autocorrelation: a stock that outperformed 10 seconds ago tends to continue briefly.

target_lag2/3/6	target.shift(2/3/6)	Extended return memory at 20s, 30s, 60s. Together with lag1, builds a short-term return momentum profile.
wap_lag1	wap.shift(1)	Price level one step ago. Enables mean-reversion detection relative to current WAP.
ofi_roll3 / ofi_roll5	ofi_ratio.rolling(3/5).mean()	3-step and 5-step smoothed OFI. Reduces tick noise while preserving the directional signal.
target_roll3_mean	target.rolling(3).mean().shift(1)	Noise-smoothed short-term trend. shift(1) is critical — without it, current target is included, causing data leakage.
target_roll5_std	target.rolling(5).std().shift(1)	Local return volatility. High values signal turbulent periods where the model should widen uncertainty intervals.
wap_velocity_lag2/5	wap_velocity.shift(2/5)	Delayed price momentum at 20s and 50s. Captures medium-term momentum decay.
ofi_ratio_lag5	ofi_ratio.shift(5)	Order flow imbalance 50 seconds ago. Medium-term OFI persistence.

4.5 Group 5 — Cross-Sectional Features

Cross-sectional features are the most important group for this problem. Because the target is a relative return — stock return minus index return — the most informative signal at any moment is how this stock's order book differs from the market average, not its absolute level.

These features are computed using `groupby([date_id, seconds_in_bucket])` — that is, across all 200 stocks at the same moment in time. This is the only group computed along the cross-sectional dimension rather than the time-series dimension.

Feature	Formula	Financial Interpretation
ofi_vs_idx	ofi_ratio – mean(ofi_ratio across stocks)	OFI relative to market average. If this stock has unusually strong buy pressure vs. the cross-sectional mean, it will outperform the index — exactly what the target measures. coef $\approx$ 0.10.
depth_vs_idx	depth_imbalance – mean(depth_imbalance)	Supply/demand tilt relative to market. Captures stock-specific order book asymmetry beyond the market-wide direction.
wap_vel_vs_idx	wap_velocity – mean(wap_velocity)	Price momentum relative to market momentum. A direct proxy for the target variable. coef = 4.48, second strongest feature after target_lag1.
cs_ofi_rank	percentile_rank(ofi_ratio within moment)	Non-parametric relative OFI. Ranks each stock from 0 to 1 among all 200 stocks at that exact second. Robust to OFI outliers and distributional shifts.

5. Feature Selection via BayesianRidge

5.1 Why Not Full ARD with NUTS?

The original plan was to use Automatic Relevance Determination (ARD) with the NUTS sampler for feature selection. ARD places a separate precision hyperparameter α_i on each weight w_i , automatically shrinking weights with no predictive signal toward zero. While this is principled Bayesian feature selection, it is computationally infeasible on 5 million rows: even with a 10,000-row subsample and 36 features, NUTS requires 30–60 minutes of sampling time due to the large number of gradient computations per step.

5.2 BayesianRidge as Practical ARD

BayesianRidge (scikit-learn) provides a practical approximation that is mathematically equivalent to ARD in the evidence maximisation (Type II Maximum Likelihood) framework. It learns separate precision hyperparameters α (weight precision) and λ (noise precision) by maximising the log marginal likelihood:

$$\log p(y | X, \alpha, \lambda) = \frac{1}{2} n \log \lambda - \frac{1}{2} \log |\alpha I + \lambda X^T X| - \frac{1}{2} \lambda \|y - X\mu_N\|^2 + \text{const}$$

Features with no predictive signal receive large α values — their weights are shrunk to near zero automatically. This is the same mechanism as ARD, implemented without MCMC sampling. A 50,000-row random subsample (approximately 1% of the full dataset) is used for computational efficiency, running in under 15 seconds.

5.3 Selection Results

Features with posterior $|\text{coef}| > 0.02$ are retained. From 28 candidates, 22 are selected and 6 are dropped. The two most important features dominate by a large margin: `target_lag1` ($\beta = 7.86$) and `wap_vel_vs_idx` ($\beta = 4.48$), reflecting short-term autocorrelation and cross-sectional momentum as the primary signals. The dropped features (`wap_rolling5_std`, `auction_active`, `wap_acceleration`, `matched_size_diff`, `matched_ratio`, `imb1`) are either redundant with retained features or carry insufficient independent signal.

6. Walk-Forward Cross-Validation

6.1 Why Standard K-Fold Fails

Standard K-fold cross-validation randomly partitions data into train and test splits. For financial time series, this approach introduces three forms of bias. First, look-ahead bias: training on future data and testing on past data allows the model to learn future patterns that would not be available in production. Second, feature leakage: lag features such as `target_lag1` computed before splitting would include future values in the training set. Third, autocorrelation leakage: adjacent observations share strong autocorrelation, so randomly shuffled splits contain information across what should be independent folds.

6.2 Purged Walk-Forward Scheme

The correct approach for financial time series is walk-forward validation: train only on past data, validate on future data, with a gap between them to prevent boundary leakage. All four windows (train, BNN, calibration, validation) are computed as contiguous blocks advancing forward in time.

A one-day gap between the training end and validation start prevents the longest lag feature (`target_lag6`, which looks 60 seconds back within a day) from seeing any validation data. All parameters scale automatically from the actual dataset size:

Parameter	Formula	Value (481 days)	Rationale
TRAIN_DAYS	$N_DAYS \times 0.65$	312 days	Sufficient history for BLR; leaves room for multiple folds
BNN_WINDOW	$TRAIN_DAYS \times 0.18$	56 days	Focuses BNN on recent dynamics; older data may represent different regimes
CAL_WINDOW	$BNN_WINDOW \times 0.17$	9 days	Held-out set for calibration scale computation
VAL_DAYS	$N_DAYS \times 0.08$	38 days	~150,000 rows per fold for reliable MAE estimation
GAP_DAYS	1 day	1 day	Prevents boundary leakage through lag features
STEP_DAYS	$= VAL_DAYS$	38 days	Non-overlapping validation sets across folds

The resulting four folds span: Train [0→311], Val [313→350]; Train [38→349], Val [351→388]; Train [76→387], Val [389→426]; Train [114→425], Val [427→464]. Each validation fold contains approximately 150,000 rows.

7. Models

7.1 Bayesian Linear Regression

7.1.1 Model Specification and Motivation

Bayesian Linear Regression differs from ordinary least squares (OLS) in one fundamental respect: it places a prior distribution on the weight vector w . OLS finds the single weight vector that minimises squared error, with no notion of uncertainty about w . BLR instead maintains a full posterior distribution over w , which becomes the source of principled uncertainty in predictions.

The BLR model assumes:

$$\begin{aligned} y &= X w + \varepsilon, & \varepsilon &\sim N(0, \lambda^{-1}) \\ w &\sim N(0, \alpha^{-1} I) \end{aligned}$$

where α is the weight precision (controls regularisation strength) and λ is the noise precision (inverse of aleatoric noise variance). Both hyperparameters are learned from data via evidence maximisation — no manual cross-validation is required. A high value of α means the prior strongly shrinks weights toward zero; a low value allows weights to grow freely. The learned α effectively determines feature importance.

7.1.2 Analytical Posterior

Under conjugate Gaussian priors, the posterior over weights is analytically tractable:

$$\begin{aligned} w \mid y, X &\sim N(\mu_N, \Sigma_N) \\ \Sigma_N &= (\alpha I + \lambda X^T X)^{-1} \\ \mu_N &= \lambda \Sigma_N X^T y \end{aligned}$$

The posterior mean μ_N is the point estimate of w , and Σ_N encodes the uncertainty. Features with low signal will have large corresponding diagonal elements in Σ_N , reflecting high uncertainty about their weights.

7.1.3 Predictive Distribution

For a new test point x^* , the posterior predictive distribution is also Gaussian:

$$\begin{aligned} \mu^* &= \mu_N^T x^* \\ \sigma^{*2} &= \lambda^{-1} + x^{*T} \Sigma_N x^* \end{aligned}$$

The variance σ^{*2} decomposes into two terms: λ^{-1} is the irreducible aleatoric noise (market randomness that no model can eliminate), and $x^{*T} \Sigma_N x^*$ is the epistemic uncertainty from imperfect knowledge of the weights. This decomposition is analytically exact — no sampling is required.

BLR is trained on the full 312-day training window (~1.2 million rows per fold). The resulting std/MAE ratio is approximately 1.74 — the model slightly overestimates uncertainty, consistent with the conservative nature of the Gaussian prior. BLR achieves mean MAE of 2.422 bp.

7.2 Monte Carlo Dropout BNN

7.2.1 The Dropout-as-Bayes Connection

Gal and Ghahramani (2016) showed that a neural network trained with dropout is mathematically equivalent to approximate variational inference in a deep Gaussian process. The key insight is that at test time, if dropout is kept active and multiple forward passes are performed with different random masks, each pass samples a different sub-network — effectively sampling from the approximate posterior over network weights.

Formally, let θ_s denote the weights of the s -th sampled sub-network. With $N = 50$ Monte Carlo samples:

$$\begin{aligned}\mu_{\text{BNN}} &= (1/50) \sum_{s=1}^{50} f(x^*; \theta_s) \\ \sigma^2_{\text{BNN}} &= (1/50) \sum_{s=1}^{50} (f(x^*; \theta_s) - \mu_{\text{BNN}})^2\end{aligned}$$

The variance across forward passes reflects disagreement between different sub-networks — the model's epistemic uncertainty about the correct prediction for x^* .

7.2.2 Architecture and Training

The BNN uses a 3-layer feed-forward architecture:

```
Layer 1: Linear(d, 128) → LayerNorm → ReLU → Dropout(0.15)
Layer 2: Linear(128, 64) → LayerNorm → ReLU → Dropout(0.15)
Layer 3: Linear(64, 32) → ReLU → Dropout(0.15) → Linear(32, 1)
```

LayerNorm is preferred over BatchNorm for financial time series because BatchNorm's batch statistics can be unstable when batch composition varies (e.g., different regimes in the same batch). The Huber loss with $\delta = 2.0$ bp is used instead of MSE to limit the influence of outlier ticks in the auction data. Training uses Adam ($\text{lr} = 5 \times 10^{-4}$), cosine annealing, early stopping (patience = 12 epochs), and weight decay 10^{-4} .

The BNN is trained on only the most recent 56 days (~220,000 rows), rather than the full 312-day window. This design choice reflects the non-stationarity of the closing auction: market microstructure dynamics shift over time

as participant composition, volatility regimes, and liquidity conditions change. A model trained on 312 days learns an average that may no longer apply in the most recent period. The 56-day window prioritises recency at the cost of sample size.

In practice, BNN MAE (2.437) is nearly identical to BLR MAE (2.422). This suggests that the 22 engineered features already linearise most of the relevant interactions, leaving little for the neural network's non-linear capacity to capture. The BNN's primary contribution is not accuracy but uncertainty diversity: its std behaves differently from BLR's, enriching the ensemble's uncertainty signal.

7.3 Inverse-Variance Ensemble and Bayesian Model Averaging

BLR and BNN produce complementary uncertainty estimates: BLR tends to overestimate uncertainty ($\text{std}/\text{MAE} \approx 1.74$) while BNN tends to underestimate it ($\text{std}/\text{MAE} \approx 0.34$). An ensemble that assigns each model weight inversely proportional to its predicted variance — i.e., proportional to its stated confidence — is a form of Bayesian Model Averaging (BMA):

$$\begin{aligned}w_i &= 1 / (\sigma_i^2 + \varepsilon) \\ \mu_{\text{ens}} &= (w_{\text{BLR}} \mu_{\text{BLR}} + w_{\text{BNN}} \mu_{\text{BNN}}) / (w_{\text{BLR}} + w_{\text{BNN}}) \\ \sigma^2_{\text{ens}} &= 1 / (w_{\text{BLR}} + w_{\text{BNN}})\end{aligned}$$

Because BNN's σ is much smaller than BLR's, $w_{\text{BNN}} \gg w_{\text{BLR}}$, and the ensemble inherits BNN's low-variance behaviour. This is the worst-of-both-worlds outcome: the ensemble has similar MAE to BLR but inherits BNN's overconfidence, yielding $\text{std}/\text{MAE} \approx 0.316$. This motivates the explicit calibration step.

8. Post-Processing

8.1 Zero-Sum Adjustment

The target imposes a structural zero-sum constraint across all stocks at each (date, second) moment. A zero-sum post-processing step enforces this:

$$\hat{y}_{\text{adj}}(i, t, s) = \hat{y}(i, t, s) - (1/N) \sum_{j=1}^N \hat{y}(j, t, s)$$

This removes any cross-sectional mean bias in predictions without affecting the relative ranking of stocks, which is what determines trading signal quality. Importantly, zero-sum adjustment does not change individual predictions relative to each other — it only removes the systematic level.

8.2 Uncertainty Calibration

A calibrated model satisfies: for predictions with stated uncertainty σ , approximately 68% of actual errors should fall within $[-\sigma, +\sigma]$, and 95% within $[-2\sigma, +2\sigma]$. The std/MAE ratio is a practical proxy: a ratio of 1.0 indicates that the mean predicted uncertainty matches the mean actual absolute error, which is the first moment of calibration.

The baseline ensemble has $\text{std/MAE} \approx 0.316$, meaning its uncertainty is $3\times$ too small. A post-hoc scale factor is computed on the calibration holdout set (last 9 training days) and applied to validation predictions:

```
scale = E[|y - μ_pred|] / E[σ_pred] ≈ 3.2
σ_cal = scale · σ_raw
```

This procedure is safe from look-ahead bias because the calibration set is always temporally before the validation set. Critically, scaling only affects σ and not μ — MAE is mathematically unchanged by calibration.

9. Baseline Pipeline Results

Model	Mean MAE	Std MAE	std/MAE ratio
BLR (312 train days)	2.4222	0.1929	1.736 (overconfident)
BNN — MC Dropout (56d)	2.4370	0.2026	0.338 (underconfident)
Ensemble (BMA)	2.4223	0.1962	0.316 (underconfident)
Ensemble + Zero-Sum	2.5227	0.2102	0.316 (unchanged)

Three observations merit detailed discussion.

BLR dominates the ensemble.

BLR achieves $\text{MAE} = 2.4222$, while the full BMA ensemble achieves $\text{MAE} = 2.4223$ — a negligible difference of 0.0001 bp. The BNN adds almost no point-prediction value. This is explained by the feature engineering: the 22 selected features already capture non-linear interactions through multiplicative terms ($\text{market_urgency} = \text{price_spread} \times \text{depth_imbalance}$), polynomial-like combinations ($\text{urgency_time_weighted}$), and ranked features (cs_ofi_rank). The residual non-linearity that the BNN might capture is insufficient to improve MAE.

Calibration is asymmetric.

BLR overestimates uncertainty ($\text{std/MAE} = 1.74$) while BNN severely underestimates it (0.34). When combined via precision weighting, the ensemble inherits the BNN's low-variance behaviour because $w_{\text{BNN}} \gg w_{\text{BLR}}$. The ensemble $\text{std/MAE} = 0.316$ is worse than either individual model. This is the key motivation for explicit calibration.

Zero-sum adjustment increases MAE.

MAE rises from 2.4223 to 2.5227 after zero-sum adjustment (+0.100 bp). This is expected: the adjustment removes the cross-sectional mean of predictions, which may have carried partial signal. The increase in MAE reflects how much of the model's apparent accuracy came from predicting market direction rather than relative cross-sectional returns. The $\text{MAE+ZS} / \text{MAE}$ ratio of $1.04\times$ in the improved pipeline is dramatically better than the $3.7\times$ ratio observed on the partial 219-day dataset, where the model was essentially predicting market direction.

10. HMM Regime Analysis

10.1 Motivation

Financial markets are non-stationary. The statistical properties of the closing auction — average order flow direction, price volatility, book depth — change depending on broader market conditions. A model trained uniformly on all days learns an average that may be suboptimal in any specific regime. Detecting latent market states and conditioning models on them is a principled approach to handling non-stationarity.

10.2 HMM Formulation

A Gaussian Hidden Markov Model is fitted to detect latent market regimes. The observation vector x_t contains four market-level aggregates at each (date, second), computed across all 200 stocks: mean OFI, OFI standard deviation, mean depth imbalance, and WAP velocity standard deviation. These are cross-sectional aggregates because regimes are market-wide phenomena — stock-level features would confuse the model.

The HMM assumes a discrete-time Markov chain of K latent states (regimes):

$$\begin{aligned} z_t \mid z_{t-1} &\sim \text{Categorical}(A_{\{z_{t-1}\}}), \quad A_{\{ij\}} = p(z_t = j \mid z_{t-1} = i) \\ x_t \mid z_t = k &\sim N(\mu_k, \Sigma_k), \quad k \in \{0, \dots, K-1\} \end{aligned}$$

Parameters (transition matrix A , emission means μ_k and covariances Σ_k , initial distribution π) are estimated via the Baum-Welch algorithm (Expectation-Maximisation for HMMs). K is selected by the Bayesian Information Criterion:

$$\text{BIC}(K) = -2 \cdot \log p(X \mid \hat{\theta}) + n_{\text{params}} \cdot \log(N)$$

10.3 Results and Regime Interpretation

BIC selects $K = 5$ regimes. The Forward-Backward algorithm provides soft regime probabilities $p(z_t = k \mid \text{all observations})$ — richer than hard Viterbi assignments because they reflect uncertainty about the current regime. Mean entropy per time step = 0.111 (max 1.233), indicating that most steps are assigned to a dominant regime with high confidence.

Regime	Frequency	ofi_mean	wap_vol	Self-Transition	Interpretation
R0	19.3%	-0.842	-0.201	0.938	Sell pressure, shallow book, uniform direction
R1	16.6%	-0.553	+1.215	0.761	Chaotic / stressed market — high price dispersion
R2	26.9%	≈0.000	-0.283	0.925	Neutral / balanced — most common normal trading day
R3	21.0%	+0.232	-0.428	0.993	Coordinated buy, very stable — most persistent regime
R4	16.3%	+1.302	-0.001	0.946	Strong buy pressure, deep book, high imbalance

The near-diagonal structure of the transition matrix indicates high regime persistence. Regime R3 (coordinated buying) is the most persistent with self-transition probability 0.993, meaning once the market enters this coordinated state it typically remains there for many consecutive days. Regime R1 (chaotic) is the least persistent (0.761), reflecting that stressed market episodes are transient.

Importantly, direct transitions between extreme regimes are nearly impossible. The probability of transitioning from R0 (strong selling) to R4 (strong buying) in a single step is 0.000 — the market must pass through intermediate states, reflecting realistic dynamics.

11. Three Improvements

11.1 Extended Lag Features

The baseline feature set includes only `target_lag1`, `target_lag6`, and `wap_lag1` as temporal memory features. Given the strong short-term autocorrelation in closing auction dynamics, this is insufficient. Eight additional features are added, all computed inside `groupby([stock_id, date_id])`:

Feature	Motivation
<code>wap_velocity_lag2</code> , <code>wap_velocity_lag5</code>	Delayed memory of price acceleration at 20s and 50s. Captures medium-term momentum decay.
<code>target_lag2</code> , <code>target_lag3</code>	Return at 20s and 30s. Completes the short-term autocorrelation profile with <code>lag1</code> .
<code>ofi_ratio_lag5</code>	Order flow imbalance 50 seconds ago. Medium-term OFI persistence.
<code>target_roll3_mean</code>	3-step rolling mean with <code>shift(1)</code> . Noise-smoothed momentum signal. <code>shift(1)</code> is critical to prevent data leakage.
<code>target_roll5_std</code>	5-step rolling std with <code>shift(1)</code> . Local return volatility estimate. High values signal where model uncertainty should widen.
<code>ofi_roll5</code>	5-step rolling OFI. Wider-window order flow trend.

After applying extended features and dropping NaN rows from lag operations, the dataset shrinks from 2,343,564 to 1,867,527 rows. These additional features reduce MAE by approximately 0.023 bp in the improved pipeline, attributable to better short-term autocorrelation capture.

11.2 Within-Fold Uncertainty Calibration

The problem is that BLR has $\text{std}/\text{MAE} \approx 1.74$ and BNN has $\text{std}/\text{MAE} \approx 0.34$, and their ensemble inherits BNN's overconfidence with $\text{std}/\text{MAE} \approx 0.316$. A well-calibrated model should have $\text{std}/\text{MAE} = 1.0$.

The calibration procedure uses the last 9 training days as a held-out calibration set, separately from the validation set. This is safe from look-ahead bias because the calibration period always precedes the validation period temporally:

```
Train [day 0 → 302] → fit BLR + BNN
Calib [day 303 → 311] → compute scale = E[|error|] / E[σ]
Val [day 313 → 350] → apply σ_cal = scale · σ_raw
```

The scale factor is approximately 3.2 across all folds, confirming that the raw predicted standard deviation is consistently 3× too small. After calibration, ensemble $\text{std}/\text{MAE} = 1.009$ across all folds — nearly exactly the target of 1.0. MAE is mathematically unchanged because calibration only scales σ , not μ .

11.3 HMM Mixture of Experts

11.3.1 Why MoE Rather Than Raw Regime Features?

A natural first approach is to add the five regime probabilities $p(z_t = k | X)$ as raw input features to the BNN. This was tried and produced only −0.005 MAE improvement. The problem is that the model must learn, from scratch, what these regime probabilities mean and how to act on them differently for each regime — a difficult inductive task.

The Mixture of Experts approach is fundamentally different: a separate BLR + BNN expert is trained for each regime, using only data from that regime. Each expert therefore learns the dynamics of its specific regime in isolation. At inference, experts are combined via the Law of Total Expectation and Total Variance.

11.3.2 Law of Total Variance

Given K regime experts, each producing predictions (μ_r, σ_r) and soft weights $p_r = p(z_t = r | X)$:

$$\begin{aligned} E[Y] &= \sum_r p_r \cdot \mu_r \\ \text{Var}[Y] &= \sum_r p_r \cdot \sigma_r^2 + \sum_r p_r \cdot (\mu_r - E[Y])^2 \end{aligned}$$

The first term is the expected within-regime variance (each expert's uncertainty, weighted). The second term is the between-expert disagreement: when different regime experts produce very different predictions, this term inflates the total uncertainty automatically. This is the key contribution of MoE — the model can express “I am uncertain because I don't know which regime we are in.”

11.3.3 Critical Implementation Details

Two implementation bugs significantly degraded MoE performance in earlier versions and required careful fixing:

Global scaler requirement:

All regime experts must share a single StandardScaler fitted on the full training data. Per-regime scalers map the same raw feature value to different normalised values in different experts — a value of `ofi_ratio` = 0.3 might normalise to 3.6 in one expert's space and 0.33 in another's. Mixing predictions from experts trained in incompatible feature spaces is incoherent. In an earlier version, per-regime scalers caused MoE MAE to collapse to 0.55, falsely appearing as an improvement but actually being meaningless.

BNN minimum sample threshold:

Regime 1 (Chaotic) is relatively rare and non-uniformly distributed across time. In some folds, the 56-day BNN window contains as few as 391 rows for Regime 1. A 3-layer BNN with 128 neurons has far more parameters than 391 training samples and will memorise rather than generalise. A minimum threshold of `MIN_BNN` = 8,400 rows was imposed: regimes below this threshold use BLR-only, without a BNN component.

12. Improved Pipeline Results

Model	Mean MAE	Std MAE	std/MAE (raw)	std/MAE (calibrated)
Ensemble (raw)	2.3997	0.1993	0.317	—
Ensemble + Cal ✓	2.3997	0.1993	0.317	1.009
MoE (raw)	2.4883	0.2228	0.491	—
MoE + Cal	2.4883	0.2228	0.491	1.154

The Ensemble + Calibration model achieves the best overall result: MAE = 2.3997 bp and std/MAE = 1.009, essentially exactly at the calibration target of 1.0. This represents a MAE improvement of 0.023 over the baseline ensemble, attributable primarily to the extended lag features.

The MoE approach yields worse MAE (+0.089 over Ensemble+Cal) and unstable calibration in folds 3 and 4 (std/MAE = 1.26 and 1.36 respectively). Regime 1 falls below the `MIN_BNN` threshold in 2 of 4 folds (391 and

7,026 rows versus the 8,400 minimum), reverting to BLR-only for that regime. The disagreement term from the Law of Total Variance (mean = 0.039) inflates std beyond what the calibration scale factor can correct. This is a clear negative result.

The mean calibration scale factors are 3.19 (Ensemble) and 2.47 (MoE), confirming that raw predicted standard deviations are dramatically smaller than actual errors in both architectures. This scale ≈ 3 is a known limitation of MC Dropout as a Bayesian approximation: it underestimates the posterior width relative to the true predictive distribution.

13. Discussion

13.1 Why BNN Does Not Beat BLR

The failure of BNN to outperform BLR in MAE is the most unexpected result of this project. Three explanations are proposed. First, the feature engineering already captures most non-linear interactions: `market_urgency` = `price_spread` × `depth_imbalance` is a product of two features; `urgency_time_weighted` = `market_urgency` × (1 + 540/stc) is a more complex non-linear combination; `cs_ofi_rank` is a non-parametric transformation. After these explicit non-linearities are introduced in feature space, the residual non-linearity for BNN to capture is limited.

Second, the 56-day BNN training window may be too short for the network to learn non-linear patterns that BLR misses with its 312-day linear fit. The BNN sees 220,000 rows across 56 days; for a 30-feature model with 128→64→32 hidden layers, this may not be sufficient to reliably learn complex patterns without overfitting.

Third, financial signals at the 10-second resolution in closing auctions may simply be largely linear. The dominant signals (`target_lag1`, `wap_vel_vs_idx`) are strongly linear features, and the residual signal may be too noisy for non-linear methods to capture reliably.

13.2 Limitations

Several limitations constrain the scope of this project. The most fundamental is architectural: every model treats each (stock, date, second) row independently. The target, however, is defined relative to the cross-sectional distribution of all stocks at each moment. A model that truly captures this structure should observe all 200 stocks simultaneously at each time step and produce joint predictions — a cross-sectional architecture not implemented here.

The calibration approach (scale factor on held-out set) is a post-hoc correction rather than a principled solution. Conformal prediction would provide distribution-free coverage guarantees without requiring Gaussian assumptions on residuals. SGLD or SWAG would provide more principled posterior sampling for the BNN, potentially producing raw std/MAE closer to 1.0 without requiring post-hoc calibration.

Finally, the HMM is fitted once on the full dataset. In production, it would need to be updated as new trading days arrive, and the regime labels would need to be estimated causally (without forward-looking information from future days).

14. Conclusions

This project presents a complete Bayesian pipeline for high-frequency equity prediction on the Optiver closing auction dataset. The pipeline combines analytically exact BLR uncertainty, approximate BNN posterior inference via MC Dropout, HMM regime detection, and principled within-fold calibration. The main findings are:

- Calibration is the most important practical contribution. std/MAE improves from 0.317 (raw ensemble) to 1.009 (calibrated ensemble) with no change to predicted means or MAE. A well-calibrated model can be directly used for risk-adjusted position sizing.
- The signal in this dataset is largely linear over 56-day windows. BNN adds negligible MAE improvement over BLR alone. The 22 engineered features, particularly cross-sectional and lag features, linearise most of the available non-linear structure.
- Cross-sectional features are the primary alpha source. The ratio MAE+ZS / MAE = 1.04× in the final pipeline (vs. 3.7× in an early partial-data version) confirms that the model correctly learns relative cross-sectional positioning rather than market direction.
- HMM identifies 5 interpretable market regimes with high persistence. The most stable regime (coordinated buying, R3) has self-transition probability 0.993. Regime detection succeeds at characterising distinct market states.
- MoE yields worse MAE (+0.089) due to Regime 1 data scarcity in some folds. This honest negative result highlights the challenge of rare regimes in walk-forward settings.
- Calibration scale ≈ 3.2 throughout, confirming a known limitation of MC Dropout as a Bayesian approximation: it systematically underestimates posterior width.